

## Chapter 5: Exploring Forces

### NCERT Class 8 Science (Curiosity)

#### 1. Probe & Ponder (In-Text Questions)

##### Q1. Why does it feel harder to pedal a bicycle when going uphill than on flat ground?

Answer: When riding a bicycle uphill, you have to work against Earth's gravitational force, which pulls both you and the bicycle downward along the slope. On flat ground, gravity acts vertically downward perpendicular to your direction of motion, so it does not directly resist forward movement. Thus, extra **muscular force** is required to pedal uphill.

##### Q2. Why is it easier to slip on a wet surface?

Answer: A thin layer of water fills the microscopic irregularities on the floor and shoe soles, drastically reducing **frictional force** (the contact force responsible for grip). With significantly lower friction, foot traction drops, making it easier to slip.

##### Q3. Why do we feel 'light' or like we are 'floating' just after our swing reaches its highest point and begins to come down?

Answer: At the peak of a swing's arc, the swing momentarily comes to a stop before reversing direction. As it accelerates downward under gravity, the upward normal reaction force exerted by the seat on your body decreases significantly, creating a momentary sensation of weightlessness or lightness.

#### 2. Q1. Match the items in Column A with the items in Column B.

| Column A (Type of Force) | Column B (Example)                                                             |
|--------------------------|--------------------------------------------------------------------------------|
| (i) Muscular force       | (b) A child lifting a school bag.                                              |
| (ii) Magnetic force      | (e) A compass needle pointing North.                                           |
| (iii) Frictional force   | (a) A cricket ball stopping on its own just before touching the boundary line. |
| (iv) Gravitational force | (c) A fruit falling from a tree.                                               |
| (v) Electrostatic force  | (d) A balloon rubbed on a woollen cloth attracts hair strands.                 |

##### Q2. State whether the following statements are True or False.

(i) A force is always required to change the speed of motion of an object.

Answer: True

Reasoning: According to the fundamental laws of motion, an object cannot change its speed (accelerate or decelerate) without the action of an unbalanced net external force.

(ii) Due to friction, the speed of the ball rolling on a flat ground increases.

Answer: False

Reasoning: Friction is an opposing contact force that acts in the direction opposite to the motion, causing the rolling ball to slow down and eventually stop.

(iii) There is no force between two charged objects placed at a small distance apart.

Answer: False

Reasoning: Charged objects exert an **electrostatic force** (a non-contact force) on each other even across a distance—repelling if they carry like charges and attracting if they carry opposite charges.

**Q3. Two balloons rubbed with a woollen cloth are brought near each other. What would happen and why?**

Answer: Observation: The two balloons will **repel each other** and move apart.

Explanation: When both balloons are rubbed with a woollen cloth, they acquire the same type of static electric charge (usually negative). Since like charges repel each other, a non-contact electrostatic force acts between them, pushing them apart.

**Q4. When you drop a coin in a glass of water, it sinks, but when you place a bigger wooden block in water, it floats. Explain.**

Answer: For the Coin: The density of the metal coin is much greater than that of water. The downward gravitational force (weight of the coin) is greater than the maximum upward **buoyant force** exerted by the displaced water, causing the coin to sink.

For the Wooden Block: Wood is less dense than water. The upward buoyant force exerted by the displaced water is sufficient to balance the downward weight of the wooden block, allowing it to float.

**Q5. If a ball is thrown upwards, it slows down, stops momentarily, and then falls back to the ground. Name the forces acting on the ball and specify their directions:**

**(i) During its upward motion:**

Gravitational force: Acts vertically downward (opposing motion, causing deceleration).

Air resistance (Friction due to air): Acts vertically downward (opposing the upward movement).

**(ii) During its downward motion:**

Gravitational force: Acts vertically downward (in the direction of motion, causing acceleration).

Air resistance: Acts vertically upward (opposing the downward movement).

**(iii) At its topmost position:**

Gravitational force: Acts vertically downward ( $W = mg$ ).

(Air resistance is zero at the instant velocity becomes zero).

**Q6. Think of a way so that when the ball is released from the same point P, it stops:**

**(i) Before the point A:**

Solution: We increase friction on the horizontal track before point A. This can be done by laying down a rough surface, such as a woollen cloth, sandpaper, or carpet, which increases the frictional force opposing the ball's rolling motion.

**(ii) After crossing the point A:**

Solution: Decrease friction on the track. This can be achieved by using a smoother surface like polished glass, plastic, or tile, or applying a lubricant (like oil or talcum powder) so the ball retains its kinetic energy longer and rolls past point A.

**Q7. Why do we sometimes slip on smooth surfaces like ice or polished floors? Explain.**

Answer: Walking requires a contact force called friction between the soles of our shoes and the ground to prevent sliding. Smooth surfaces like ice or freshly polished floors have very few surface irregularities, offering minimal friction. Without adequate friction to provide forward grip, our feet lose traction, causing us to slip.

**Q8. Is any force being applied to an object in a non-uniform motion?**

Answer: Yes. Non-uniform motion means an object's speed, direction of motion, or both are continuously changing over time. Any change in the state of motion (acceleration or deceleration) requires a non-zero net external force acting on the object.

**Q9. The weight of an object on the Moon becomes one-sixth of its weight on the Earth. What causes this change? Does the mass of the object also become one-sixth of its mass on the Earth?**

Reason for weight change: Weight ( $W$ ) is the gravitational pull exerted on an object by a celestial body:

$$W = m \cdot g$$

Because the Moon has a much smaller mass than Earth, its surface gravitational acceleration ( $g_{\text{moon}}$ ) is roughly  $\frac{1}{6}$  of Earth's gravity ( $g_{\text{earth}}$ ).

Consequently, an object's weight on the Moon is one-sixth of its weight on Earth.

Does mass change? No. Mass ( $m$ ) is the fundamental quantity of matter contained within an object and remains constant everywhere in the universe, regardless of location or local gravitational pull.

**Q10. Three objects 1, 2, and 3 of the same size and shape but made of different materials are placed in water. They dip to different depths. If the weights of the three objects 1, 2, and 3 are  $w_1$ ,  $w_2$ , and  $w_3$  respectively, then:**

\* (i)  $w_1 = w_2 = w_3$

\* (ii)  $w_1 > w_2 > w_3$

\* (iii)  $w_2 > w_3 > w_1$

\* (iv)  $w_3 > w_1 > w_2$

Correct Answer: (ii)  $w_1 > w_2 > w_3$

Detailed Explanation:

For a floating object, the upward buoyant force ( $F_b$ ) exerted by the liquid must balance the object's downward weight ( $W$ ):

$$F_b = w$$

By Archimedes' Principle, the buoyant force equals the weight of the fluid displaced by the submerged portion of the object. Object 1 dips deepest into the water, displacing the greatest volume of water, and thus requires the largest upward buoyant force to remain afloat. Therefore, Object 1 is the heaviest ( $w_1$ ), while Object 3 submerges the least and is the lightest ( $w_3$ ). Hence:

$$w_1 > w_2 > w_3$$